Individual Weekly Report – Week 6

Feature Engineering

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Department	Artificial Intelligence and Data Science
Project Title	Youth Smoking and Drug Consumption: From Raw Data to Visual Insights
Role	Data Visualization and Dashboard Development
Week	Week 6 – Feature Engineering

Objective of the Week

The aim of this week was to show the effect of feature engineering through visualization. When features are scaled or transformed, graphs help to show what has actually changed in the data.

Work Done

Integrating before-and-after graphs: I added before-and-after graphs into the visualization workflow. Placing the two views next to each other makes it easy to see the effect of each transformation.

Comparing feature distributions: I compared the distribution of the features before and after transformation. This showed how the shape and range of the data changed after the feature engineering steps.

Visualizing scaled and transformed data: I prepared visual comparisons of the scaled and transformed data. These comparisons helped to confirm that the changes made by the team worked as expected.

Improving the presentation of results: I improved the way the feature engineering results were shown, by using clearer labels, titles and layouts, so that the comparisons are easy for anyone to read.

Tools and Techniques Used

Before-and-after comparison graphs and distribution plots for scaled and transformed features.

Outcome of the Week

The team could see the effect of feature engineering clearly. The comparisons also gave useful visual material for the project documentation.

What I Learned

I learned that showing the same data before and after a change is a simple but powerful way to explain what a data processing step does.

Contribution to the Project

My comparison graphs helped the team check and explain the feature engineering work.

Challenges Faced

It was difficult to show the changes in the data clearly when the scale of the features was different before and after transformation. I handled this by placing the graphs side by side and using clear titles and labels.

Plan for Next Week

In Week 7, I will prepare PCA visualizations and plan how the dimensionality reduction results will be shown in the dashboard.

Weekly Summary

Week 6 showed how visualization can explain data processing steps. The before-and-after graphs made the effect of feature engineering easy to understand.